Security in the Cloud TASK – 3

1. **Enable Multi-Factor Authentication (MFA) on your personal AWS or Azure cloud account using the built-in authenticator app option, and take a screenshot of the MFA setup confirmation page.**

   Ans :

   - Sign in to the AWS Management Console using your personal AWS account.
   - Open the IAM service:
     - Search for IAM in the AWS Console search bar.
     - Open IAM (Identity and Access Management).
   - Open your account security settings:
     - Click your account name in the top-right corner.
     - Select Security credentials.
   - Locate the Multi-factor authentication (MFA) section.
   - Click Assign MFA device (or Activate MFA).

2. **Simulate a login to your cloud account from a new browser or incognito mode and document the steps and prompts you receive with MFA enabled.**

   Ans :

   - Open a new browser window or open an Incognito/Private browsing window.
   - Go to the AWS sign-in page:
     - Select Sign in to the AWS Management Console.

    - Choose Root user (for a personal account) or IAM user (if using an IAM account).
- Enter your AWS account credentials:
    - Enter the registered email address (root user) or IAM user name.
    - Click Next.
    - Enter the account password.
    - Click Sign in.
- Open your authenticator app on your mobile device.
- Find the AWS MFA entry and enter the current 6-digit verification code displayed in the app.
- Click Submit or Verify MFA.

3. **Disable MFA temporarily on your cloud account, attempt to log in again, and note the difference in the login experience and security prompts.

Hint: Compare the number of steps and any warnings shown with and without MFA.**

   Ans :

- Sign in to the AWS Management Console.
- Open the IAM service.
- Click your account name in the top-right corner and select Security credentials.
- Find the Multi-factor authentication (MFA) section.
- Select the active MFA device.
- Choose Deactivate MFA (or remove the MFA device if required).

- Confirm the action to temporarily disable MFA.
- Test Login Without MFA
- Open a new Incognito/Private browser window.
- Go to the AWS sign-in page.
- Enter your AWS account username/email.
- Enter your password.
- Observe the login process:
    - AWS signs you in after password verification.
    - No MFA code prompt appears.
    - Fewer authentication steps are required.

4. **Write a short scenario (3-5 lines) describing how an attacker might try to access a cloud account without MFA, and explain how enabling MFA would block or alert you to this attempt.**

Ans : An attacker may obtain a user's cloud account password through phishing or a data breach and try to log in without MFA enabled.
Without MFA, the attacker could access the account using only the stolen password.
With MFA enabled, the attacker would also need the verification code from the user's authenticator app, blocking the login attempt.
The user would notice an unexpected MFA request or login alert, indicating a possible unauthorized access attempt.